

DACHENG ZHAO

Electrical Engineering Student

San Diego, CA | University of California, San Diego | B.S. Electrical Engineering

SUMMARY

Electrical Engineering student focused on building a strong foundation in electronics, embedded systems, programming, and hardware design. Experienced in turning engineering concepts into practical projects and exploring the interaction between hardware and software, with interests spanning embedded systems, electronics, hardware design, robotics, and computer engineering.

EDUCATION

University of California, San Diego

2025 – Present

B.S. in Electrical Engineering

- Studying electrical engineering with an emphasis on circuits, electronics, embedded systems, and computer engineering.

University of California, Davis

Previous Study

Electrical Engineering Coursework

- Completed foundational coursework in mathematics, circuits, electronics, and engineering.

PROJECTS

Line-Following Robot

ESP32 · C/C++ · Sensors · PID

- Designed and programmed a sensor-based robot using an ESP32.
- Implemented weighted sensor readings and PID-based steering for line tracking.

Analog Amplifier Design

Op-Amps · Circuit Analysis · Electronics

- Designed and analyzed op-amp amplifier circuits, including gain stages, filters, buffers, and signal-conditioning circuits.

Robot Combat & 3D Design

Fusion 360 · 3D Design · Robotics · Fabrication

- Gained hands-on experience with mechanical and 3D design using Fusion 360 while participating in the Northern California Robot Combat Competition.
- Combined physical design, fabrication, electronics, and engineering problem-solving.

PCB & Hardware Design

PCB Design · Electronics · Embedded Systems

- Developed experience designing electronic hardware and PCBs.
- Connected circuit concepts with practical physical layouts and embedded-system applications.

Personal Portfolio

HTML · CSS · Responsive Design

- Designed and developed a responsive personal resume website to present academic experience, technical skills, and engineering projects.

SKILLS

Programming

C / Arduino, Python, MATLAB

Hardware & Design

Circuit Analysis, Analog Electronics, Op-Amps, PCB Design, 3D Design, Fusion 360

Embedded Systems

ESP32, Arduino, Microcontrollers, PID Control, Motor Control

Tools

Git, Figma, Linux, Windows